

Sulphides

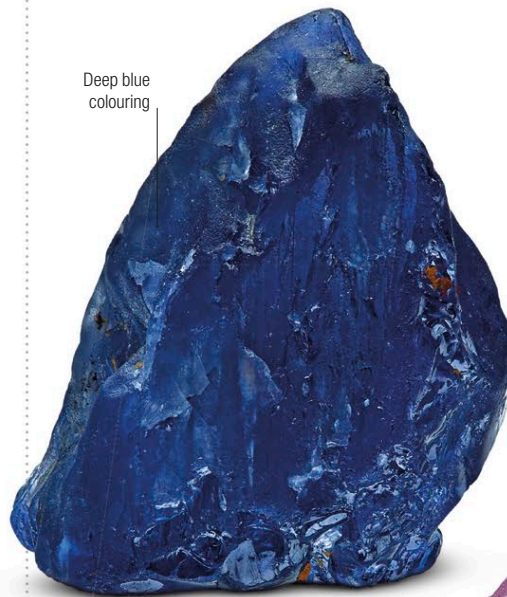
Sulphide minerals are those in which sulphur is combined with one or more metals. Many of the sulphides are brilliantly coloured, and most have low hardness and high specific gravity. Examples of sulphides include pyrite, marcasite, and sphalerite.



Sphalerite rough

Oxides

Minerals of the oxide group consist of oxygen atoms combined with a metal or semi-metal. An example of this is aluminium oxide, or corundum – ruby and sapphire. Other gemstone varieties include spinel – often mistaken for ruby – hematite, and rutile.



Deep blue colouring

Sapphire rough

Phosphates

These minerals are grouped according to the similarity of their crystal structures – phosphate minerals contain phosphorus and oxygen combined in a 1:4 ratio. Some examples of phosphates that occur as gemstones include amblygonite, apatite, and turquoise.



Vitreous to pearly luster

Amblygonite rough

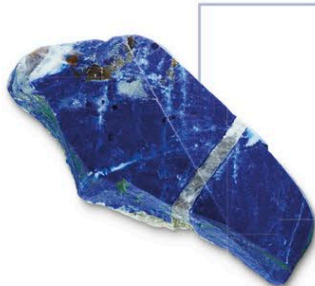


Violet colour caused by iron and natural radiation

Amethyst rough

Silicates

All silicates consist of silicon and oxygen atoms, structured as a central silicon atom with oxygen atoms around it in various configurations. Silicates are divided into subgroups according to the varying structural configurations of their atoms; of these, inosilicates are sub-divided into two further groups, as below. Silicates include many gemstones such as quartz and tourmaline.



Tectosilicates

Tectosilicates include lazurite (above), opal, quartz varieties such as amethyst, and more.



Phyllosilicates

This group includes chrysocolla, soapstone, and clay minerals, among others.



Single-chain inosilicates

Single chain inosilicates include kunzite (above), enstatite, diopside, and others.



Double-chain inosilicates

Double-chain inosilicates include nephrite (above), edenite, and others.



Cyclosilicates

Emerald (above) is the best-known member of this group, which features tourmaline.



Sorosilicates

Sorosilicates include vesuvianite, zoisite, and other minerals.